Moto armonico

$$\begin{cases} x(t+T) = x(t) \\ v(t+T) = \vec{v}(t) \end{cases}$$

T = periodo del moto

ω = pulsazione

$$x(t) = A\cos(\omega t + \varphi) \qquad A>0$$

φ = fase iniziale

$\omega t + \varphi$ = la fase del moto

$$T = \frac{2\pi}{\omega} \qquad x(t+T) = x(t) \qquad \nu = \frac{1}{T}$$

ν: mi "muovo in greco"

$$x(t+T) = A\cos[\omega(t+T)+\varphi] = A\cos(\omega t + \varphi + \omega T)$$
$$= A\cos\left(\omega t + \varphi + \omega\frac{2\pi}{\omega}\right) = A\cos(\omega t + \varphi + 2\pi$$
$$= A\cos(\omega t + \varphi) = x(t)$$

$\cos(x + 2\pi) = \cos x$

$$\Rightarrow x(t+T) = x(t)$$

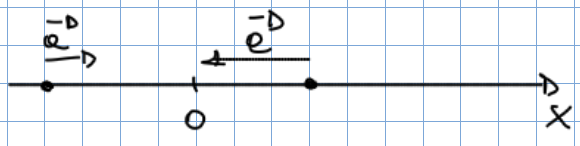
$$v(t) = \frac{dx}{dt} = \frac{d}{dt}[A\cos(\omega t + \varphi)] = A\frac{d}{dt}[\cos(\omega t + \varphi)]$$
$$= A[-\sin(\omega t + \varphi)\omega] = -A\omega\sin(\omega t + \varphi)$$

$$a(t) = \frac{dv}{dt} = -\omega A\frac{d}{dt}[\sin(\omega t + \varphi)] = -\omega^2 A\cos(\omega t + \varphi)$$

$$\begin{cases} -\omega^2 A\cos(\omega t + \varphi) \\ x(t) = A\cos(\omega t + \varphi) \end{cases} \Rightarrow a(t) = -\omega^2 x(t)$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

eq differenziale del moto armonico



essendo $a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

$-1 \leq \cos(\omega t + \varphi) \leq 1$

$A \leq x(t) \leq A$

$-\omega A \leq v(t) \leq \omega A$

$-\omega^2 A \leq a(t) \leq \omega^2 A$

$v(t) = -\omega A \sin(\omega t + \varphi)$

$$\left.\begin{aligned} x^2 &= A^2 \cos^2(\omega t + \varphi) \\ \left(\frac{v}{\omega}\right)^2 &= A^2 \sin^2(\omega t + \varphi) \end{aligned}\right\} \Rightarrow x^2 + \frac{v^2}{\omega^2} = A^2\left[\cos^2(\omega t + \varphi) + \sin^2(\omega t + \varphi)\right] = A^2$$

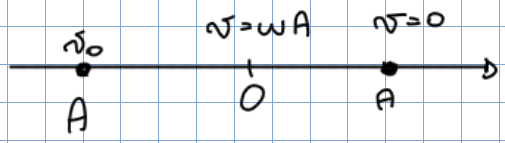
$$\Downarrow$$

$$x = A \Rightarrow v = 0$$

$$x = 0 \Rightarrow v^2 = \omega^2 A^2 \Rightarrow v = \omega A$$

$$\Downarrow$$

$$x^2 + \frac{v^2}{\omega^2} = A^2 \Rightarrow v^2 = \omega^2(A^2 - x^2)$$



$x(t) = A\cos(\omega t + \varphi)$

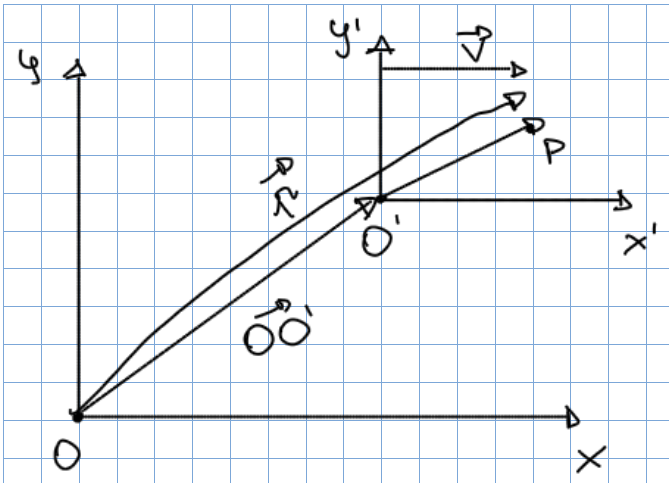
condizione iniziale

$$\begin{cases} x(0) = x_0 \\ v(0) = 0 \end{cases} \Rightarrow \begin{cases} x_0 = A\cos\varphi \\ 0 = -\omega A \sin\varphi \end{cases} \Rightarrow \sin\varphi = 0 \Rightarrow \varphi = 0;\pi$$

$\varphi = 0 \Rightarrow x_0 = A > 0 \Rightarrow x(t) = x_0\cos(\omega t)$

$\varphi = \pi \Rightarrow x_0 = A\cos\pi = -A < 0 \Rightarrow x_0 < 0 \Rightarrow x(t) = -x_0\cos(\omega t + \pi)$

$$x(t) = x_0\cos(\omega t)$$

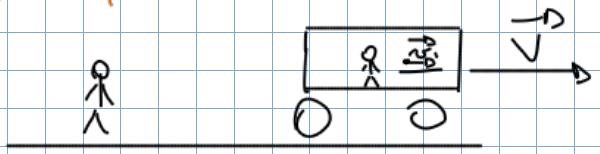


$$\vec{r} = \vec{r}' + \vec{OO'} \qquad \vec{v}' = \frac{d\vec{P}}{dt}$$

$$\vec{v}_{O'} = \frac{d\vec{OO'}}{dt}$$

$$\frac{d\vec{r}}{dt} = \frac{d\vec{P}}{dt} + \frac{d\vec{OO'}}{dt} \Rightarrow \boxed{\vec{v} = \vec{v}' + \vec{v}_{O'}}$$

Esempio



$v' = 2\ m/s$

$V = 10\ m/s$

$\vec{v} = 10 + 2 = 12\ m/s$

Cosa succede alle accelerazioni?

$$\vec{a} = \frac{d\vec{v}}{dt}, \quad \vec{a}' = \frac{d\vec{v}'}{dt}, \quad \vec{A} = \frac{d\vec{v}_{O'}}{dt}$$

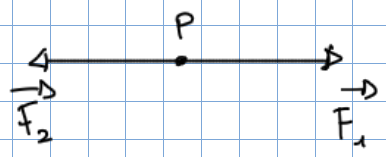
$$\frac{d\vec{v}}{dt} = \frac{d\vec{v}'}{dt} + \frac{d\vec{v}_{O'}}{dt} \Rightarrow \boxed{\vec{a} = \vec{A} + \vec{a}'}$$

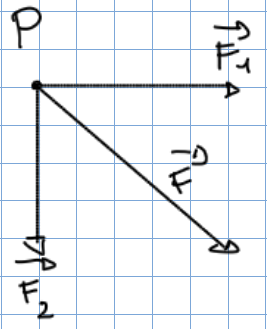
$\vec{v}_{O'}$ costante $\Rightarrow \vec{A} = 0 \Rightarrow \boxed{\vec{a} = \vec{a}'}$

LA FORZA

se un punto materiale accelera diciamo che su quel corpo incide una forza

$\vec{a} = \vec{a} + \vec{A}$





$\vec{F} = \vec{F}_1 + \vec{F}_2$

$\sum \vec{F}_i = 0$ → se vera il punto materiale è detto libero

Sistema inerziale (primo principio delle dinamiche)

$\sum_i \vec{F}_i = 0 \Rightarrow \vec{a} = 0 \Rightarrow \vec{v} =$ costante

$$\begin{cases} \vec{v} = \vec{V} + \vec{v}' \\ a = \vec{V} + \vec{a}' \end{cases}$$

$\vec{V} =$ costante $\Rightarrow \vec{A} = 0 \Rightarrow \vec{a} = \vec{a}' \Rightarrow \vec{a}' = 0$

Secondo principio delle dinamiche

$$\vec{F} = m_i \cdot \vec{a}$$

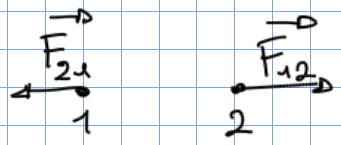
massa inerziale

$F = kg \frac{m}{s^2} = N$

$\vec{a} = \frac{\vec{F}}{m_i}$

$$\vec{a} = \frac{d\vec{v}}{dt} \cdot \frac{d^2\vec{r}}{dt} \rightarrow \boxed{\vec{F} = m\frac{d^2\vec{r}}{dt} = F\left(\vec{r}, \vec{v}, t\right)} \Rightarrow \vec{r} = \vec{r}(t)$$

terzo principio della dinamica



$$\vec{F}_{21} = -\vec{F}_{12}$$

l'azione e la reazione sono applicate in punti diversi